# Track and overhead line against limits

One LiDAR scan, one curve of the track. The gauge of the rails and the height and stagger of the contact wire are read along the chainage and compared with reference limits.

Run run1 · gps_pointcloud_00.laz · 15,924,732 points · 264.2 m · reference line 032f505d2216 · pipeline 461f18528f0e · page made 2026-10-14

Gauge | Contact wire

| -2.7 mm | 1428.3 mm | 1435.8 mm | 103 | 60 % |
|---|---|---|---|---|
| **mean gauge, from the nominal 1435.0 mm** | **tightest point shown** | **widest point shown** | **points beyond PLAN** | **of points within PLAN** |
| 1432.3 mm · spread ± 1.4 mm · 260 points | -3.8 mm against the PLAN minimum of 1432.0 mm | +9.2 mm below the PLAN maximum of 1445.0 mm | PLAN 103 · UH1 46 · UH2 17 · KRIT 0 · 23 stretches | 157 of 260 points |

## Settings · Gauge

Gauge: the distance between the inner faces of the two rails, nominally 1435 mm, read from the LiDAR points in every 25 cm slab.

REFERENCE LIMITS

Trafikverket H4, 160 < V ≤ 200 km/h, point faults: PLAN 1432.0–1445.0 · UH1 1431.0–1447.0 · UH2 1430.0–1455.0 · KRIT 1428.0–1463.0 mm

POINTS OF THE CHART

1 m stretches (mean of 4 slabs)

IGNORE STRETCHES SHORTER THAN

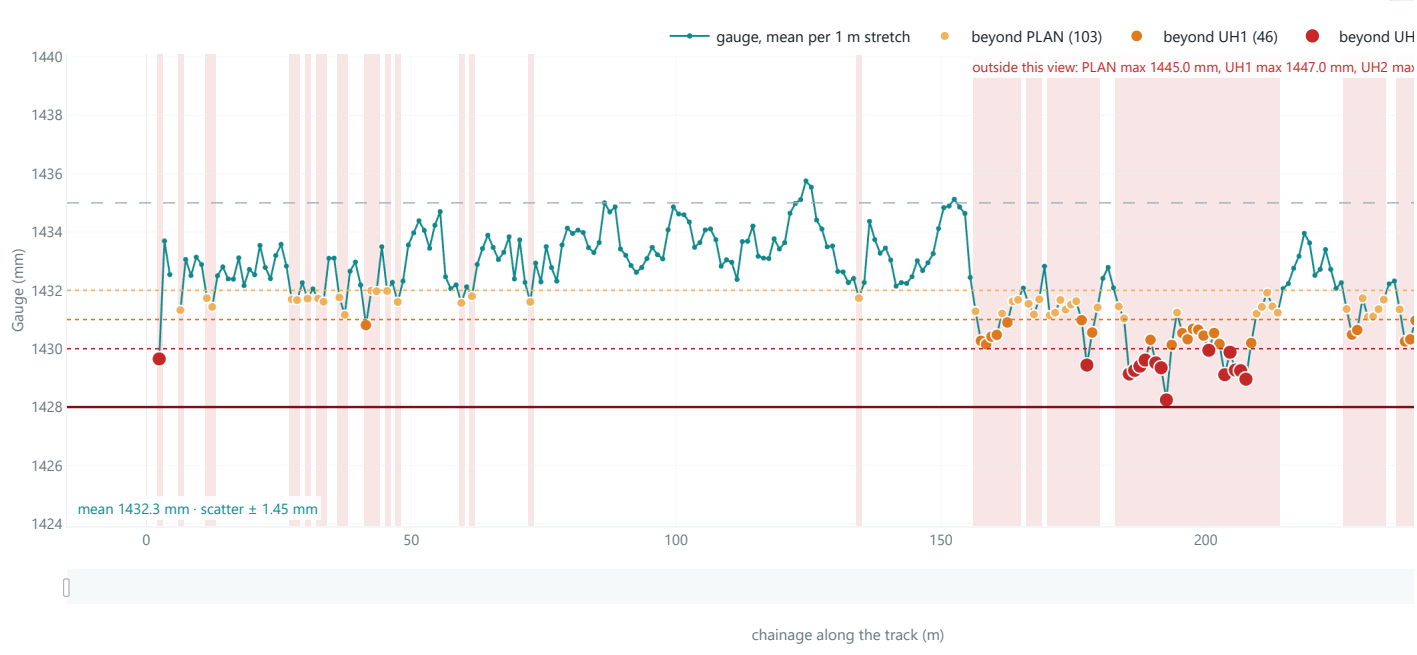
1 m

VERTICAL SCALE

fit the track

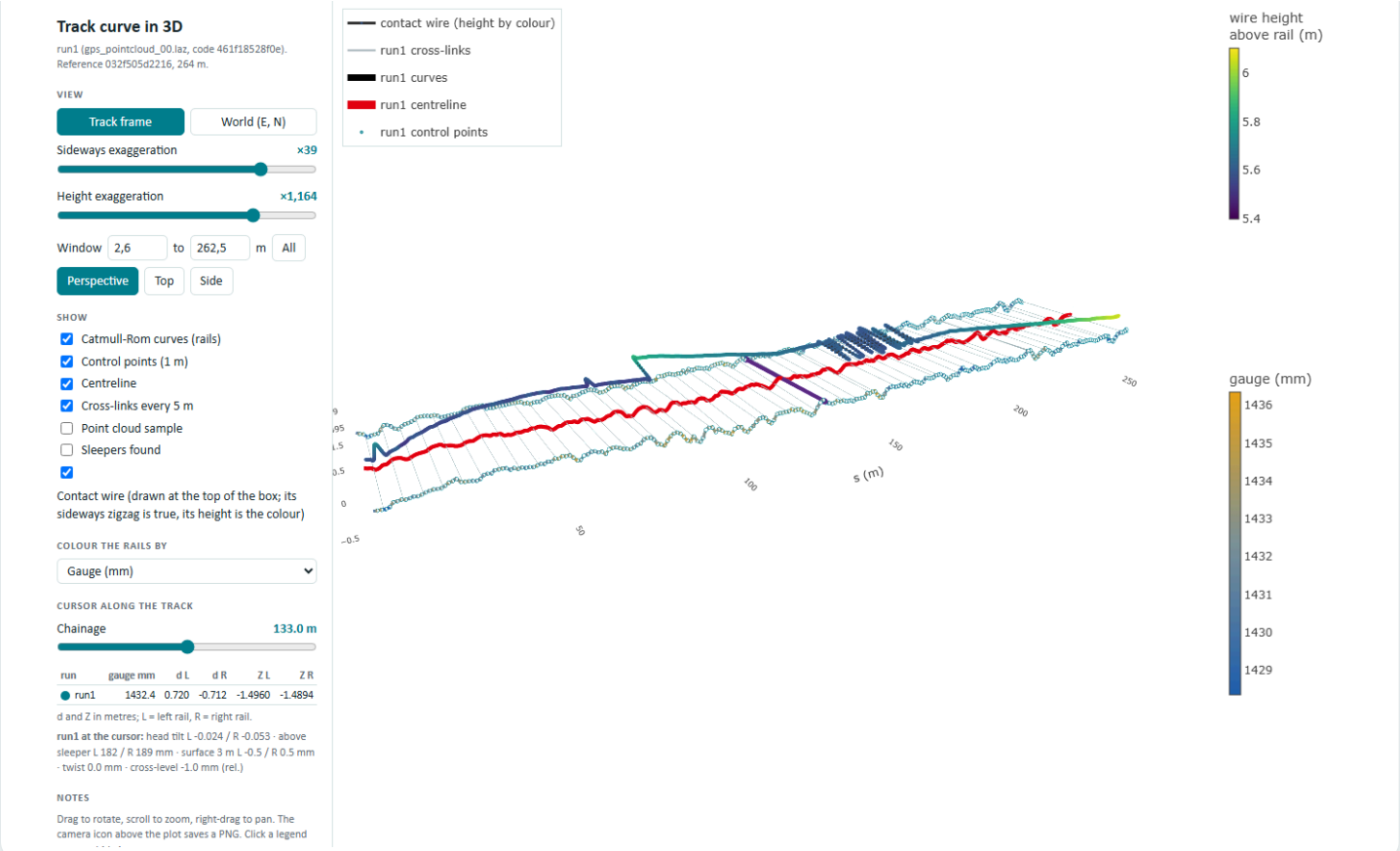
## Gauge along the chainage

Teal: gauge, mean per 1 m stretch. Limit lines: Trafikverket H4, 160 < V ≤ 200 km/h, point faults, from PLAN (dotted, light) to KRIT (solid, dark). Dashed grey: the nominal value. Larger markers are coloured by the most severe limit level a point exceeds; the legend counts them. Red shading: a stretch beyond a limit. Drag below the chart to zoom; double-click to reset.




## The curve in 3D

The run as a 3D curve of both rails. Set "Colour the rails by" to gauge to see tight and wide stretches on the track itself; tick "Contact wire" under Show to draw the wire along the top of the box (its sideways zigzag is true, its height is the colour); the Window and Cursor of the viewer use the same chainage as the chart. Open it in its own tab.




Still image of the 3D view (perspective, rails coloured by gauge, contact wire on); the live view is in monitor.html.

## Flagged stretches

23 stretches of gauge beyond a level of the Trafikverket H4, 160 < V ≤ 200 km/h, point faults limits; the most severe level reached is named.

| KIND | FROM | TO | LENGTH | EXTREME VALUE | FROM NOMINAL | FROM THE MEAN |
|---|---|---|---|---|---|---|
| beyond UH2 | 2.5 m | 2.5 m | 1.0 m | 1429.7 mm | -5.3 mm | -2.6 mm |
| beyond PLAN | 6.5 m | 6.5 m | 1.0 m | 1431.3 mm | -3.7 mm | -1.0 mm |
| beyond PLAN | 11.5 m | 12.5 m | 2.0 m | 1431.4 mm | -3.6 mm | -0.9 mm |
| beyond PLAN | 27.5 m | 28.5 m | 2.0 m | 1431.7 mm | -3.3 mm | -0.6 mm |
| beyond PLAN | 30.5 m | 30.5 m | 1.0 m | 1431.7 mm | -3.3 mm | -0.6 mm |
| beyond PLAN | 32.5 m | 33.5 m | 2.0 m | 1431.6 mm | -3.4 mm | -0.7 mm |
| beyond PLAN | 36.5 m | 37.5 m | 2.0 m | 1431.2 mm | -3.8 mm | -1.1 mm |
| beyond UH1 | 41.5 m | 43.5 m | 3.0 m | 1430.8 mm | -4.2 mm | -1.5 mm |
| beyond PLAN | 45.5 m | 45.5 m | 1.0 m | 1432.0 mm | -3.0 mm | -0.3 mm |
| beyond PLAN | 47.5 m | 47.5 m | 1.0 m | 1431.6 mm | -3.4 mm | -0.7 mm |

## How to read this: gauge

**Gauge** is the distance between the inner faces of the two rails, nominally 1435 mm, read from the LiDAR points in every 25 cm slab. Height is not used, so the sensor-relative height of the scan does not matter here.

**The limits** are Trafikverket's requirements for track geometry, TRVINFRA-00013 version 2.0 (2021-10-11), tables K11.2 and K11.3: the deviation of a single gauge reading from 1435 mm, and the 100 m sliding mean, per speed class H0 to H5 and in four levels. PLAN is the planning limit, UH1 and UH2 the two maintenance limits (UH2 is a safety requirement: repair before the next inspection), and KRIT the critical limit where action is immediate. The scan lies on Södra stambanan south-west of Höör (km 566.2 to 566.4, 180 km/h), so class H4 (160 to 200 km/h) is preselected; it can be changed in the menu.

**Selected now: Trafikverket H4, 160 < V ≤ 200 km/h, point faults.** deviation of a single reading from 1435 mm, measured unfiltered 14 mm below the rail top; four levels, PLAN, UH1, UH2 and KRIT. **PLAN** (min 1432.0, max 1445.0 mm): planning limit (alert limit): plan the work. **UH1** (min 1431.0, max 1447.0 mm): maintenance limit 1: repair soon, before UH2 is reached. **UH2** (min 1430.0, max 1455.0 mm): maintenance limit 2, a safety requirement: repair before the next inspection. **KRIT** (min 1428.0, max 1463.0 mm): critical limit: act at once.

**Stretches, not points.** A fault is a stretch, so the page can average the points over 1, 2 or 5 m and drops flagged stretches shorter than the chosen minimum. The point-to-point noise of the raw series is 0.68 mm; a point of this chart averages 4 of them, so its own noise is about 0.34 mm. A single point a little past a limit may

therefore be noise; a stretch of many points cannot be.

**This run.** Mean 1432.3 mm, scatter ± 1.4 mm, from 1428.3 to 1435.8 mm. Points beyond each level of Trafikverket H4, 160 < V ≤ 200 km/h, point faults: PLAN 103, UH1 46, UH2 17, KRIT 0. The scan is one pass; between two passes a positioning error would come on top of what is shown here.

Data: output/runs/run1/. Page made by monitor.py.

# Track and overhead line against limits

One LiDAR scan, one curve of the track. The gauge of the rails and the height and stagger of the contact wire are read along the chainage and compared with reference limits.

Run run1 · gps_pointcloud_00.laz · 15,924,732 points · 264.2 m · reference line 032f505d2216 · pipeline 461f18528f0e · page made 2026-10-14

Gauge | **Contact wire**

| +0.179 m | 57 | 77 % | -0.001 m | 48 | 81 % |
|---|---|---|---|---|---|
| **mean wire height, from the nominal 5.500 m** | **wire height: points beyond the limits** | **wire height: within the limits** | **mean wire stagger, from the nominal 0.000 m** | **wire stagger: points beyond the limits** | **wire stagger: within the limits** |
| 5.679 m · from 5.519 to 6.051 m · 249 points | TSI Energy, nominal range · 2 stretches | 192 of 249 points | -0.001 m · from -0.581 to 0.588 m · 249 points | TSI Energy, 1600 mm pantograph · 2 stretches | 201 of 249 points |

## Settings · Contact wire height above the rail

Contact wire height above the rail: the height of the overhead contact wire above the plane of the two rail heads, measured every metre from the points of the lowest wire above the track (contact_wire.py); the pantograph must find the wire within a height window.

REFERENCE LIMITS: TSI Energy, nominal range: nominal range 5.000–5.750 m

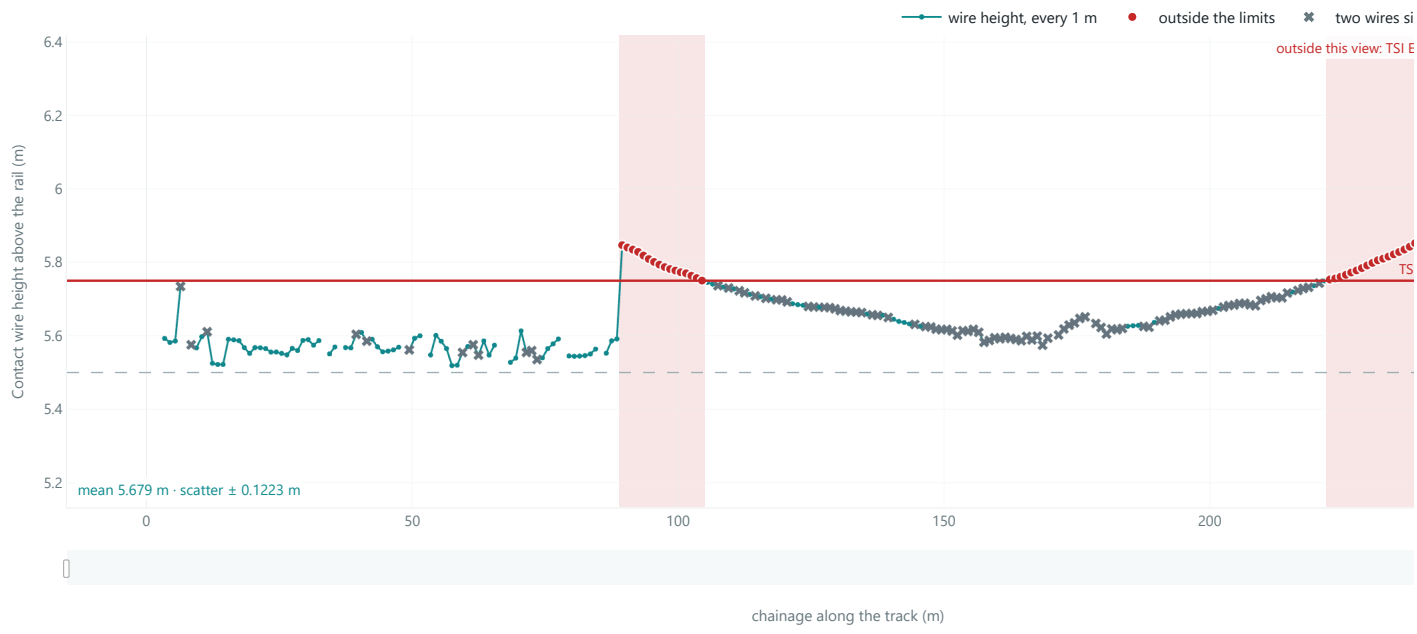
POINTS OF THE CHART: every 1 m (as measured)

IGNORE STRETCHES SHORTER THAN: 1 m

VERTICAL SCALE: fit the track

## Contact wire height above the rail along the chainage

Teal: wire height, every 1 m. Red lines: the TSI Energy, nominal range limits. Dashed grey: the nominal value. Larger markers are coloured by the most severe limit level a point exceeds; the legend counts them. Red shading: a stretch beyond a limit. Grey crosses: two wires side by side (overlap) or a fitting. Drag below the chart to zoom; double-click to reset.




## The rails and the contact wire in 3D

Track frame: chainage along, sideways across, height up (true metres, the along-track axis compressed). The wire zigzags over the track centre and rides about 5.6 m above the rails; its colour is its height. Drag to turn, scroll to zoom.

left rail | right rail | track centre | contact wire

### Flagged stretches

2 stretches of contact wire height above the rail beyond a level of the TSI Energy, nominal range limits; the most severe level reached is named.

| KIND | FROM | TO | LENGTH | EXTREME VALUE | FROM NOMINAL | FROM THE MEAN |
|---|---|---|---|---|---|---|
| outside the limits | 89.5 m | 104.5 m | 16.0 m | 5.847 m | +0.347 m | +0.168 m |
| outside the limits | 222.5 m | 262.5 m | 41.0 m | 6.051 m | +0.551 m | +0.372 m |

### How to read this: contact wire height above the rail

**Contact wire height above the rail** is the height of the overhead contact wire above the plane of the two rail heads, measured every metre from the points of the lowest wire above the track (contact_wire.py); the pantograph must find the wire within a height window. It is measured against the rails in the same cross-section, so the sensor-relative height of the scan cancels out.

**The limits** are the height values of the European TSI for the energy subsystem (Regulation (EU) 1301/2014, point 4.2.9.1): a nominal height between 5.00 and 5.75 m and at most 6.50 m with tolerances and pantograph uplift. The Swedish nominal height is 5.50 m; the Swedish maximum is to be confirmed with the supervisor.

**Selected now: TSI Energy, nominal range.** 4.2.9.1: nominal height 5.00 to 5.75 m. **nominal range** (min 5.000, max 5.750 m): outside the nominal range of the TSI.

**Stretches, not points.** A fault is a stretch, so the page can average the points over 1, 2 or 5 m and drops flagged stretches shorter than the chosen minimum. The point-to-point noise of the raw series is 0.0169 m; a point of this chart carries all of it. A single point a little past a limit may therefore be noise; a stretch of many points cannot be.

**This run.** Mean 5.679 m, scatter ± 0.122 m, from 5.519 to 6.051 m. 57 points lie outside the TSI Energy, nominal range limits. The scan is one pass; between two passes a positioning error would come on top of what is shown here.

Data: output/runs/run1/. Page made by monitor.py.

### Settings · Contact wire stagger

Contact wire stagger: the sideways position of the contact wire relative to the track centreline (positive to the left): the zigzag that spreads the wear over the pantograph strip; too far out and the wire leaves the strip.

REFERENCE LIMITS: TSI Energy, 1600 mm pantograph: maximum -0.400–0.400 m

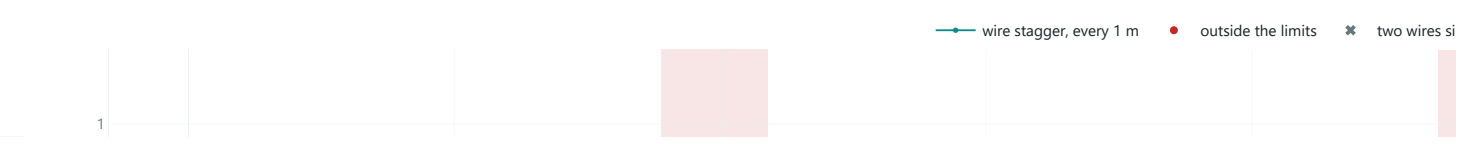
POINTS OF THE CHART: every 1 m (as measured)

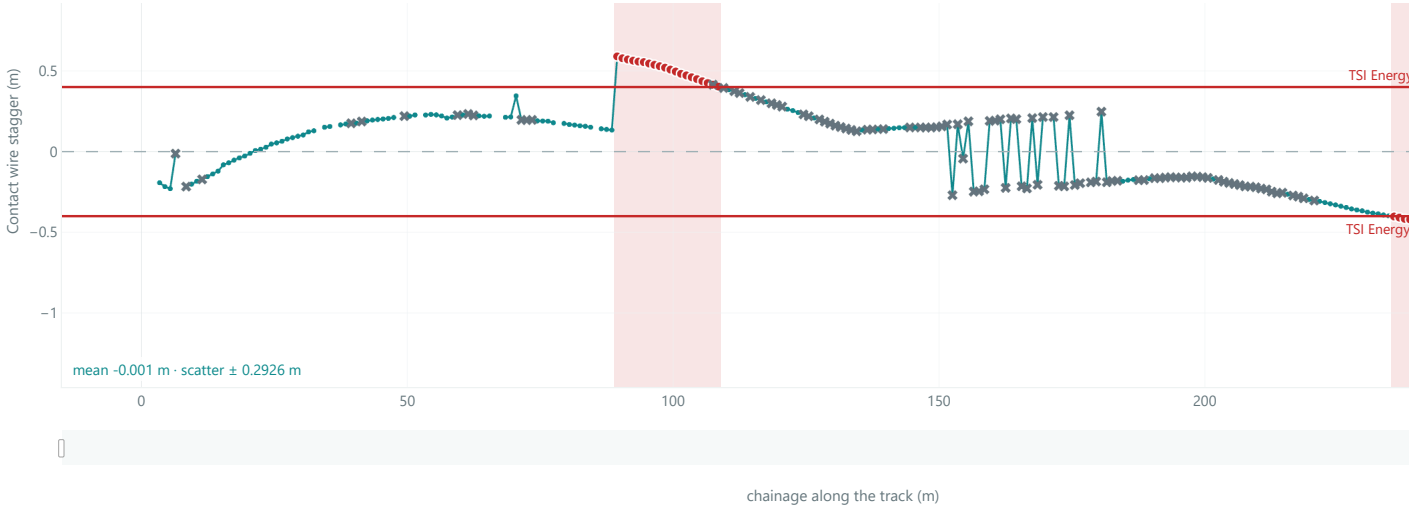
IGNORE STRETCHES SHORTER THAN: 1 m

VERTICAL SCALE: fit the track

### Contact wire stagger along the chainage

Teal: wire stagger, every 1 m. Red lines: the TSI Energy, 1600 mm pantograph limits. Dashed grey: the nominal value. Larger markers are coloured by the most severe limit level a point exceeds; the legend counts them. Red shading: a stretch beyond a limit. Grey crosses: two wires side by side (overlap) or a fitting. Drag below the chart to zoom; double-click to reset.






## Flagged stretches

2 stretches of contact wire stagger beyond a level of the TSI Energy, 1600 mm pantograph limits; the most severe level reached is named.

| KIND | FROM | TO | LENGTH | EXTREME VALUE | FROM NOMINAL | FROM THE MEAN |
|---|---|---|---|---|---|---|
| outside the limits | 89.5 m | 108.5 m | 20.0 m | 0.588 m | +0.588 m | +0.589 m |
| outside the limits | 235.5 m | 262.5 m | 28.0 m | -0.581 m | -0.581 m | -0.580 m |

## How to read this: contact wire stagger

**Contact wire stagger** is the sideways position of the contact wire relative to the track centreline (positive to the left): the zigzag that spreads the wear over the pantograph strip; too far out and the wire leaves the strip. It is measured against the rails in the same cross-section, so the sensor-relative height of the scan cancels out.

**The limits** are the maximum lateral deviation of the contact wire from the track centre in the European TSI for the energy subsystem (Regulation (EU) 1301/2014, table 4.2.9.2): 0.40 m for a 1600 mm pantograph head and 0.55 m for a 1950 mm head, under cross wind. The scan shows the wire at rest, so the wind allowance is on top. The value for Swedish 1800 mm heads is to be confirmed.

**Selected now: TSI Energy, 1600 mm pantograph.** table 4.2.9.2: at most 0.40 m from the track centre under cross wind. **maximum** (min -0.400, max 0.400 m): further from the centre than a 1600 mm head allows.

**Stretches, not points.** A fault is a stretch, so the page can average the points over 1, 2 or 5 m and drops flagged stretches shorter than the chosen minimum. The point-to-point noise of the raw series is 0.0816 m; a point of this chart carries all of it. A single point a little past a limit may therefore be noise; a stretch of many points cannot be.

**This run.** Mean -0.001 m, scatter ± 0.293 m, from -0.581 to 0.588 m. 48 points lie outside the TSI Energy, 1600 mm pantograph limits. The scan is one pass; between two passes a positioning error would come on top of what is shown here.

Data: output/runs/run1/. Page made by monitor.py.